

Diffusion Model Mathematics Notes

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1 Diffusion Model Derivation

1.1 Master Equations

De-noising Diffusion Probabilistic Model (DDPM) is an application of stochastic Markov process, derived from the following continuous time VP-SDE (Variance-Preserving Stochastic Differential Equation):

$$d\mathbf{X}(t) = -\frac{\beta(t)}{2}\mathbf{X}(t)dt + \sqrt{\beta(t)}d\mathbf{W}(t) \quad (1)$$

$\mathbf{X}(t)$ is the stochastic random variable (n-dim, in general), by definition $d\mathbf{X}(t) \equiv \mathbf{X}(t+dt) - \mathbf{X}(t)$.

$\mathbf{W}(t)$ is the standard Wiener process, by definition $d\mathbf{W}(t) \equiv \mathbf{W}(t+dt) - \mathbf{W}(t) = \sqrt{dt}\epsilon$, where ϵ is the Gaussian random number sampled from normal distribution with mean=0 and variance=1.

$\beta(t) \geq 0$, for $t \geq 0$, is the variance scheduler function.

This SDE is known as the time-dependent **Ornstein-Uhlenbeck** process in physics, where the 1st-term in the right-hand-side of SDE is the drift term, and 2nd-term is the diffusion term. Physically it was to describe the particle velocity distribution under a drift force and a fluctuation (noise) force.

The solution of $\mathbf{X}(t)$, or equivalently the distribution function $q(\mathbf{X}(t))$, can be found by solving the corresponding **Fokker-Planck equation** (aka **Kolmogorov forward equation**), which is the equation to describe the time evolution of conditional probability $q(x_t|x_s), t > s \geq 0$:

$$\frac{\partial}{\partial t}q(x_t|x_s) = \frac{\beta(t)}{2} \left(\frac{\partial}{\partial x_t}(x_t q(x_t|x_s)) + \frac{\partial^2 q(x_t|x_s)}{\partial x_t^2} \right); t > s \geq 0 \quad (2)$$

Here we use the short notation for conditional probability density function:

$$q(\mathbf{X}(t) = x; t = t | \mathbf{X}(s) = y; t = s) \equiv q(x; t | y; s) \equiv q(x_t|x_s) \quad (3)$$

note that $\mathbf{X}(t)$ is the stochastic random variable, and x, y are the sampled results at time t, s , for $t > s$.

1.2 Analytic Solution

The solution can be found by changing the variables:

$$B(t) = \int_0^t \beta(\tau) d\tau \quad (4)$$

$$\alpha_t \equiv \alpha(t) = \exp(-B(t)) \quad (5)$$

$$y_t = x_t \exp(B(t)/2) \quad (6)$$

The analytic solution is:

$$q(x_t|x_s) = \exp \left(-\frac{(x_t - \sqrt{\alpha_t/\alpha_s}x_s)^2}{2(1 - \alpha_t/\alpha_s)} \right) \quad (7)$$

This is the Gaussian distribution, here we drop the normalization constant.

By definition, $\alpha(t=0) = 1$, so:

$$q(x_t|x_0) = \exp \left(-\frac{(x_t - \sqrt{\alpha_t}x_0)^2}{2(1 - \alpha_t)} \right) \quad (8)$$

1.2.1 forward sampling

For the forward sampling (noising process) at arbitrary time $t > 0$:

$$x_t = \sqrt{\alpha_t}x_0 + \sqrt{1 - \alpha_t}\epsilon_t \quad (9)$$

where ϵ_t is the Gaussian noise. For a given initial data x_0 and a pre-defined signal decay function $\alpha(t)$, one can compute "noised data" at arbitrary time t using forward sampling formula.

1.2.2 signal decay function α_t

The signal decay function α_t (called signal power in some other literatures) controls how the initial data (input image) lost its signature with gradually increasing noise.

Two common types of α_t are:

linear scheduler(original DDPM paper)

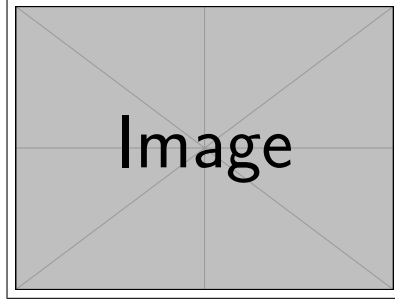
$$\beta(t) = \frac{1}{T} \left(\beta_0 + (\beta_1 - \beta_0) \frac{t}{T} \right); 0 \leq t \leq T \quad (10)$$

$$\alpha_t = \exp \left(- \int_0^t \beta(\tau) d\tau \right) \quad (11)$$

cosine scheduler(proposed in iDDPM paper):

$$\alpha_t = \cos^2 \left(\frac{\pi}{2} \frac{t}{T} \right); 0 \leq t \leq T \quad (12)$$

Figure 1: signal decay function α_t



1.2.3 reverse sampling

Ideally if we know the reverse conditional probability $q(x_s|x_t)(s < t)$, we are able to apply the reverse sampling in the reverse diffusion process, and back to original initial data x_0 . However, the conditional probability $q(x_s|x_t)$ is unknown unless the marginal probability $q(x_0)$ of the initial data distribution is given. Unfortunately, $q(x_0)$ is usually unknown and is what we want to model.

The key step of DDPM is to use $q(x_s|x_t, x_0)$ to approximate the $q(x_s|x_t)$, where $q(x_s|x_t, x_0)$, the probability of sampling x_s at time $s < t$ when conditioned on x_t and x_0 , can be calculated analytically by (for $t > s > 0$) using Bayes' rule:

$$q(x_s|x_t, x_0) = \frac{q(x_t, x_s, x_0)}{q(x_t, x_0)} = \frac{q(x_t|x_s, x_0)q(x_s|x_0)q(x_0)}{q(x_t|x_0)q(x_0)} = \frac{q(x_t|x_s)q(x_s|x_0)}{q(x_t|x_0)} \quad (13)$$

$$q(x_s|x_t, x_0) = \exp \left(- \frac{(x_s - \mu_s(t, x_t, x_0))^2}{2\Sigma_{s,t}^2} \right) \quad (14)$$

$$\mu_s(t, x_t, x_0) = \frac{1 - \alpha_s}{1 - \alpha_t} \sqrt{\frac{\alpha_t}{\alpha_s}} x_t + \frac{1 - \alpha_t/\alpha_s}{1 - \alpha_t} \sqrt{\alpha_s} x_0 \quad (15)$$

$$\Sigma_{s,t}^2 = \frac{1 - \alpha_s}{1 - \alpha_t} \left(1 - \frac{\alpha_t}{\alpha_s} \right) \quad (16)$$

This is also a Gaussian distribution with mean μ_s and variance $\Sigma_{s,t}^2$,

1.2.4 DDPM reverse sampling formula

$$x_0^\theta(x_t) = \frac{x_t - \sqrt{1 - \alpha_t} \epsilon_t^\theta(x_t)}{\sqrt{\alpha_t}} \quad (17)$$

$$x_s = \frac{1 - \alpha_s}{1 - \alpha_t} \sqrt{\frac{\alpha_t}{\alpha_s}} x_t + \frac{1 - \alpha_t/\alpha_s}{1 - \alpha_t} \sqrt{\alpha_s} x_0^\theta(x_t) + \Sigma_{s,t} \epsilon \quad (18)$$

1.2.5 DDIM reverse sampling formula

$$x_0^\theta(x_t) = \frac{x_t - \sqrt{1 - \alpha_t} \epsilon_t^\theta(x_t)}{\sqrt{\alpha_t}} \quad (19)$$

$$x_s = \sqrt{\alpha_s} x_0^\theta(x_t) + \sqrt{1 - \alpha_s - \eta \Sigma_{s,t}^2} \epsilon_t^\theta(x_t) + \eta \Sigma_{s,t} \epsilon \quad (20)$$

$$\eta = 1, \text{ mathematically identical to DDPM sampling} \quad (21)$$

$$\eta = 0, \text{ deterministic reverse sampling} \quad (22)$$